

PAPER_387 — Relativistic SCm Velocity Parameter Update: $v_{SCm} = 0.99c$

Source: grok_share_cfdcad2f5.txt, lines ~9600–10122 (main.cpp global constants)

Section: Star Magic_construction file_040ct2025.docx — global parameter declarations

Session: 106 (grok_share_cfdcad2f5.txt full analysis)

CP4 Class: vSCmRelativisticParameterUpdateCalculator (CP4 #38)

Abstract

This paper presents a UQFF analysis of Relativistic SCm Velocity Parameter Update: $v_{SCm} = 0.99c$, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

Prior UQFF implementations used a preliminary value $v_{SCm} = 1 \times 10^8$ m/s for the Superconductive medium velocity parameter in the reactive energy term E_{react} . The Star Magic_construction file_040ct2025.docx Grok thread formalizes an updated value:

$$v_{SCm} = 0.99 \times c = 0.99 \times 2.998 \times 10^8 \text{ m/s} = 2.968 \times 10^8 \text{ m/s}$$

This represents the first formal assignment of v_{SCm} to a relativistic speed grounded in observational evidence from the J1610+1811 quasar jet ($z=3.122$, covered in PAPER_374).

PAPER_374 identified $v=0.99c$ as the jet velocity for J1610+1811. PAPER_375 used this in the UQFF wormhole/Meissner advanced integration context. However, **neither paper formally updated the v_{SCm} constant in the core parameter set** — this paper makes that explicit and calculates the cascading impact on the E_{react} term.

2. The v_{SCm} Parameter

2.1 Definition

v_{SCm} is the characteristic velocity of the Superconductive medium (SCm) phase in UQFF. It appears in the reactive energy term connecting quantum vacuum density to the SCm-plasma coupling:

$$E_{react} = \frac{\rho_{SCm} \cdot v_{SCm}^2}{\rho_A} \cdot e^{-\kappa t}$$

Where: - ρ_{SCm} = SCm vacuum density (kg/m^3) - v_{SCm} = SCm characteristic velocity (m/s) - ρ_A = ambient density (kg/m^3), default $\rho_A = 1 \times 10^{-23} \text{ kg/m}^3$ - κ = decay constant, default $\kappa = 0.0005 \text{ day}^{-1}$ - t = time elapsed

2.2 Parameter Update: Old vs New

Parameter	Old Value	New Value	Ratio
v_{SCm}	$1 \times 10^8 \text{ m/s}$	$2.968 \times 10^8 \text{ m/s (0.99c)}$	$2.968 \times$
v_{SCm}^2	$1 \times 10^{16} \text{ m}^2/\text{s}^2$	$8.808 \times 10^{16} \text{ m}^2/\text{s}^2$	8.808 ×

The velocity-squared amplification factor is **8.808×**, meaning all Ereact calculations using the prior value are underestimated by approximately one order of magnitude.

3. Observational Basis

The update is validated by the J1610+1811 relativistic quasar jet:

- **System:** J1610+1811, $z=3.122$ (lookback time ~ 11.5 Gyr)
- **Jet power:** $P_{\text{jet}} \approx 4 \times 10^{45}$ W
- **Jet velocity:** $v = 0.99c$
- **Lorentz factor:** $\gamma = (1 - v^2/c^2)^{-1/2} \approx 7.089$
- **Source documents:** Star Magic_09Sept2025.docx, Star Magic_construction file_04Oct2025.docx

This system demonstrates that SCm-driven plasma jets reach $0.99c$, making this the physically motivated upper-bound velocity for the SCm velocity parameter.

4. Updated Ereact Calculation

For the canonical SGR1745 parameters: - $\rho_{\text{SCm}} \approx \rho_A = 1 \times 10^{-23}$ kg/m³ - $t = 0$ (initial condition)

Old:

$$E_{\text{react}}^{\text{old}} = \frac{(1 \times 10^{-23})(1 \times 10^{16})}{1 \times 10^{-23}} \cdot e^0 = 1 \times 10^{16} \text{ J/m}^3$$

New:

$$E_{\text{react}}^{\text{new}} = \frac{(1 \times 10^{-23})(8.808 \times 10^{16})}{1 \times 10^{-23}} \cdot e^0 = 8.808 \times 10^{16} \text{ J/m}^3$$

The reactive energy increases by a factor of **8.808×** across all systems.

5. Global Constants Context (Oct 2025)

The full confirmed global parameter set from main.cpp (Oct 2025 build):

```
const double c = 2.998e8;           // Speed of light (m/s)
double v_SCm = 0.99 * c;           // SCm velocity = 2.968e8 m/s ← UPDATED
double Omega_g = 7.3e-16;          // Galactic angular velocity (rad/s)
double Mbh = 8.15e36;               // SMBH mass (kg)
double dg = 2.55e20;                // Galaxy scale distance (m)
double rho_A = 1e-23;               // Ambient density (kg/m³)
double rho_sw = 8e-21;              // Solar wind density (kg/m³)
double v_sw = 5e5;                  // Solar wind velocity (m/s)
double kappa = 0.0005;              // UQFF decay constant (day⁻¹)
double alpha = 0.001;               // Coupling constant α
double gamma = 0.00005;             // Coupling constant γ
double k1=1.5, k2=1.2, k3=1.8, k4=2.0; // MUGE layer weights
double beta_i = 0.603;              // Buoyancy coupling (≈0.6 calibrated)
double rho_v = 6e-27;               // Vacuum energy density (kg/m³)
double C_concentration = 1.0;       // Concentration factor
```

```
double f_feedback = 0.1;           // Feedback fraction
const double num_strings = 1e9;    // String count
```

6. Implications for UQFF Pipeline

6.1 Affected Equations

1. **Ereact term:** All systems using v_{SCm^2} scaling see $8.808\times$ amplification
2. **Compressed MUGE:** The v_{SCm^2}/c^2 relativistic correction factor changes from 0.1111 (old) to 0.9801 (new) — approaching unity
3. **Lorentz correction:** With $v=0.99c$, the Lorentz factor $\gamma=7.089$ is now accessible for relativistic corrections in jet-class systems

6.2 Calibration Compatibility

The calibrated constant $\kappa=0.0005/\text{day}$ (from PAPER_341) remains unchanged — the decay envelope is independent of the velocity amplitude.

The calibrated $\beta_i \approx 0.603$ (PAPER_375) also remains valid as it governs buoyancy coupling, not the SCm velocity channel.

7. Validation Cross-Reference

Reference	Connection
PAPER_374	J1610+1811 jet physics providing the $v=0.99c$ basis
PAPER_375	UQFF Wormhole/Meissner integration using $\gamma=7.089$
PAPER_341	$\kappa=0.0005/\text{day}$ calibration (unchanged by this update)
PAPER_372	Compressed MUGE 8-term base (Ereact channel)

8. Canonical Value (All Future Implementations)

$v_{\text{SCm}} = 0.99c = 2.968 \times 10^8 \text{ m/s}$ is the canonical Superconductive medium velocity.

All UQFF Python and C++ implementations should use:

```
c = 2.998e8 # m/s
v_SCm = 0.99 * c # = 2.968e8 m/s

const double c = 2.998e8;
double v_SCm = 0.99 * c; // = 2.968e8 m/s
```

Discovery Class: Parameter Formalization — First explicit canonical assignment of $v_{\text{SCm}}=0.99c$

Distinct from: PAPER_374 (J1610 jet observational context); PAPER_375 (Meissner/wormhole use of γ)

Impact: $8.808\times$ amplification of all Ereact-channel UQFF calculations